Enterprise IT SLA Intelligence Dashboard

Project description:

The Enterprise IT SLA Intelligence Dashboard is an AI-powered tool designed for corporate IT support teams to monitor and manage service tickets effectively. It uses machine learning to analyse incoming tickets and predict the likelihood of SLA (Service Level Agreement) breaches in advance. By identifying high-risk tickets early, the system helps teams take proactive action to resolve issues before deadlines are missed. This improves response efficiency, reduces SLA violations, and shifts IT support from a reactive approach to a proactive service management model.

Key Benefits:

- Real-time SLA breach prediction
- Automated incident classification and routing
- Early risk detection and proactive alerts
- Improved SLA compliance and service quality
- Enhanced operational efficiency and resource utilization

Machine Learning Workflow

- Data Collection: Captures incident details, priority, and timestamps.
- Feature Engineering: Calculates resolution time and extracts time-based features.
- Data Processing: Applies preprocessing and feature selection using saved pipeline models.
- Prediction: Generates SLA breach risk scores using the trained machine learning model.

Business Benefits

- Reduces SLA breaches and associated operational costs.
- Enables proactive resource allocation for high-risk incidents.
- Improves service quality and helpdesk performance monitoring.
- Accelerates incident resolution and reduces downtime.
- Enhances decision-making through real-time risk insights.

Technology

- **Frontend:** Streamlit
- **Backend:** Python
- **Data Processing:** Pandas, Datetime
- **Machine Learning:** Scikit-learn
- **Model Deployment:** Joblib

Challenges Overcome

- Implemented automated incident routing to ensure tickets are assigned to the correct teams without manual delay.
- Used real-time ticket duration calculations to improve the accuracy of SLA breach predictions.
- Transformed predictive risk outputs into clear, actionable recommendations using priority-based remediation plans.
- Integrated predictive analytics into existing IT incident management workflows for smoother operations.
- Improved overall operational efficiency by reducing manual effort and enabling faster response to critical tickets.
